STRING FUNCTIONS

🧠 **1. CONCAT() — Join strings**

📌 **Definition**

Combines multiple strings into one.

✅ **Syntax**

CONCAT(str1, str2, ...)

✅ **Example**

SELECT CONCAT('John', ' ', 'Doe') AS full_name;

👉 Output: John Doe

💡 **Real Use**

SELECT CONCAT(full_name, ' - ', city) FROM string_practice;

⚠️ **Important**

If any value is NULL → result becomes NULL

👉 Fix:

SELECT CONCAT(IFNULL(name,''), ' ', IFNULL(city,''))

🧠 **2. LENGTH() — String size**

📌 **Definition**

Returns number of **bytes** in a string

✅ **Example**

SELECT LENGTH('Hello');

👉 Output: 5

⚠️ **Important**

- For multibyte characters → length differs
  👉 Use:

CHAR_LENGTH('Hello')

💡 **Real Use**

SELECT full_name, LENGTH(full_name)

FROM string_practice;

🧠 **3. TRIM() — Remove spaces**

📌 **Definition**

Removes spaces from **both sides**

✅ **Example**

SELECT TRIM(' hello ');

👉 Output: hello

💡 **Custom trim**

SELECT TRIM('x' FROM 'xxxhelloxxx');

👉 Output: hello

🧠 **4. LTRIM() — Left trim**

📌 **Definition**

Removes spaces from **left side only**

✅ **Example**

SELECT LTRIM(' hello');

👉 Output: hello

🧠 5. RTRIM() — Right trim

📌 **Definition**

Removes spaces from **right side only**

✅ **Example**

SELECT RTRIM('hello ');

👉 Output: hello

💡 **Real Use Case**

Cleaning imported data:

SELECT TRIM(full_name) FROM string_practice;

🧠 6. REPLACE() — Replace text

📌 **Definition**

Replaces all occurrences of a substring

✅ **Syntax**

REPLACE(original, old, new)

✅ **Example**

SELECT REPLACE('hello world', 'world', 'SQL');

👉 Output: hello SQL

💡 **Real Use**

SELECT REPLACE(email, '@gmail.com', '')

FROM string_practice;

🧠 7. LOCATE() — Find position

📌 **Definition**

Returns position of substring

✅ **Syntax**

LOCATE(substring, string)

✅ **Example**

SELECT LOCATE('world', 'hello world');

👉 Output: 7

💡 **Real Use**

Find position of @ in email:

SELECT LOCATE('@', email)

FROM string_practice;

⚠️ **Important**

- If not found → returns 0

🧠 **8. SUBSTRING() — Extract part of string**

📌 **Definition**

Extracts a portion of a string

✅ **Syntax**

SUBSTRING(string, start, length)

✅ **Example**

SELECT SUBSTRING('Hello World', 1, 5);

👉 Output: Hello

💡 **Real Use**

SELECT SUBSTRING(phone, 1, 3)

FROM string_practice;

🔥 **9. SUBSTRING_INDEX() (Very Powerful)**

📌 **Definition**

Splits string based on delimiter

✅ **Example**

```
SELECT SUBSTRING_INDEX('john@gmail.com', '@', 1);
```

👉 Output: john

```
SELECT SUBSTRING_INDEX('john@gmail.com', '@', -1);
```

👉 Output: gmail.com

💡 **Real Use**

Email parsing, product codes, logs

🧠 **COMBINING FUNCTIONS (REAL POWER)**

🔥 **Example 1: Extract username**

```
SELECT SUBSTRING_INDEX(email, '@', 1)
FROM string_practice;
```

🔥 **Example 2: Proper case name**

```
SELECT CONCAT(
   UPPER(LEFT(full_name,1)),
   LOWER(SUBSTRING(full_name,2))
)
FROM string_practice;
```

🔥 **Example 3: Clean data**

```
SELECT TRIM(LOWER(full_name))
```

FROM string_practice;

💥 INTERVIEW-LEVEL INSIGHTS

⚡ 1. TRIM is critical in real data

Dirty data often has:

- extra spaces
- inconsistent formatting

⚡ 2. SUBSTRING + LOCATE = dynamic extraction

SELECT SUBSTRING(email, 1, LOCATE('@', email)-1)

⚡ 3. REPLACE is useful for cleaning

- remove symbols
- standardize text

⚡ 4. Always combine functions

Real queries are never single-function

🎯 FINAL SUMMARY

Function	Use
CONCAT	join strings
LENGTH	measure size
TRIM	clean spaces
REPLACE	substitute text
LOCATE	find position
SUBSTRING	extract part
SUBSTRING_INDEX	split strings

Below are **15 tough string problems with answers and clear explanations in one place.**

🔥 **1. Extract Username (NO SUBSTRING_INDEX)**

❓ **Problem**

From email, extract username (john.doe@gmail.com → john.doe)

✅ **Solution**

```
SELECT SUBSTRING(email, 1, LOCATE('@', email) - 1) AS username
FROM string_practice;
```

🧠 **Explanation**

- LOCATE('@') → finds position of @
- -1 → excludes it
- SUBSTRING → extracts left part

🔥 **2. Extract Domain Name (gmail)**

❓ **Problem**

john.doe@gmail.com → gmail

✅ **Solution**

```
SELECT
SUBSTRING(
  SUBSTRING(email, LOCATE('@', email) + 1),
  1,
  LOCATE('.', SUBSTRING(email, LOCATE('@', email) + 1)) - 1
) AS domain
FROM string_practice;
```

🧠 **Explanation**

- Extract gmail.com
- Then cut before .

🔥 **3. Capitalize Full Name**

❓ **Problem**

john doe → John Doe

✅ **Solution (2-word case)**

```
SELECT CONCAT(
  UPPER(LEFT(full_name,1)),
  LOWER(SUBSTRING(full_name,2, LOCATE(' ', full_name)-2)),
  ' ',
  UPPER(SUBSTRING(full_name, LOCATE(' ', full_name)+1,1)),
  LOWER(SUBSTRING(full_name, LOCATE(' ', full_name)+2))
) AS proper_name
FROM string_practice;
```

🧠 **Explanation**

- Break into first + last name
- Apply UPPER + LOWER

🔥 **4. Reverse String (Concept)**

❓ **Problem**

hello → olleh

✅ **Solution**

```
SELECT REVERSE(full_name) FROM string_practice;
```

🧠 **Note**

- Without REVERSE() → complex (recursive logic)
- In interview: explain approach

🔥 **5. Invalid Emails**

❓ **Problem**

Find emails not in valid format

✅ **Solution**

```
SELECT *
FROM string_practice
```

```
WHERE email NOT LIKE '%@%.%';
```

🧠 **Explanation**

- Must contain @ and .

🔥 **6. Extract Product Number (middle part)**

❓ **Problem**

prd-001-ny → 001

✅ **Solution**

```
SELECT
SUBSTRING(
  product_code,
  LOCATE('-', product_code) + 1,
  LOCATE('-', product_code, LOCATE('-', product_code)+1)
  - LOCATE('-', product_code) - 1
) AS product_num
FROM string_practice;
```

🧠 **Explanation**

- Locate first -
- Locate second -
- Extract between

🔥 **7. Count Words**

❓ **Problem**

"this is sql" → 3

✅ **Solution**

```
SELECT
(LENGTH(comments) - LENGTH(REPLACE(comments, ' ', '')) + 1) AS word_count
FROM string_practice;
```

🧠 **Explanation**

- Count spaces → words = spaces + 1

🔥 **8. Remove Numbers**

❓ **Problem**

abc123 → abc

✅ **Solution**

```
SELECT REGEXP_REPLACE(comments, '[0-9]', '')
FROM string_practice;
```

🧠 **Explanation**

- [0-9] → matches digits

🔥 **9. Find Duplicate Characters (Concept)**

❓ **Problem**

programming → r, g, m

✅ **Approach**

- Break string into characters
- GROUP BY character
- HAVING COUNT > 1

👉 SQL alone → complex (needs recursion or helper table)

🔥 **10. Remove Extra Spaces**

❓ **Problem**

hello world → hello world

✅ **Solution**

```
SELECT REGEXP_REPLACE(comments, ' +', ' ')
FROM string_practice;
```

🔥 **11. Mask Email**

❓ **Problem**

john@gmail.com → j***@gmail.com

✅ **Solution**

```
SELECT CONCAT(
  LEFT(email,1),
  '***',
  SUBSTRING(email, LOCATE('@', email))
) AS masked_email
FROM string_practice;
```

🔥 **12. First & Last Name**

❓ **Problem**

Split full name

✅ **Solution**

```
SELECT
SUBSTRING(full_name, 1, LOCATE(' ', full_name)-1) AS first_name,
SUBSTRING(full_name, LOCATE(' ', full_name)+1) AS last_name
FROM string_practice;
```

🔥 **13. Longest Name**

❓ **Problem**

Find longest name

✅ **Solution**

```
SELECT full_name
FROM string_practice
ORDER BY LENGTH(full_name) DESC
LIMIT 1;
```

🔥 **14. Initials**

❓ **Problem**

john doe → JD

✅ **Solution**

```
SELECT CONCAT(
```

```
    UPPER(LEFT(full_name,1)),
    UPPER(SUBSTRING(full_name, LOCATE(' ', full_name)+1,1))
) AS initials
FROM string_practice;
```

🔥 **15. Palindrome**

❓ **Problem**

Find palindrome names

✅ **Solution**

```
SELECT full_name
FROM string_practice
WHERE full_name = REVERSE(full_name);
```

💥 **FINAL PATTERNS (Very Important)**

🧠 **Learn these → you solve 80% problems:**

- LOCATE + SUBSTRING → dynamic extraction
- LENGTH - REPLACE → counting
- REGEXP → cleaning
- CONCAT → formatting